

1. Administrative & Study Identification

Full Study Title: Real World Outcomes and Treatment Patterns in Brazilian Patients With Hormone Receptor Positive / HER2-negative Unresectable or Metastatic Breast Cancer **Short Title/Acronym:** REALNESS (Observational study on the treatment of patients with metastatic breast cancer) **Protocol Number:** D9673R00053 **NCT Number:** NCT06676436 **Sponsor Name:** AstraZeneca **Investigational Product:** CDK4/6 inhibitors (CDK4/6i) - Real-World Evidence tracking **Phase of Development:** Observational / N/A (Real-World Evidence) **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the treatment patterns and outcomes in individuals with hormone receptor-positive (HR+) and HER2-negative unresectable or metastatic breast cancer (mBC) in Brazil. **Secondary Objectives:** To describe patient characteristics, treatment sequencing, reasons for permanent discontinuation of CDK4/6 inhibitors (due to disease progression, death, or any other cause), and overall real-world effectiveness.

2.2 Background & Rationale To generate Real-World Evidence (RWE) regarding the management of HR+/HER2- mBC in Brazil. While clinical trials show the efficacy of CDK4/6 inhibitors in this population, this retrospective study evaluates how these therapies are being used in routine Brazilian clinical practice, the sequencing of treatments, and real-world patient outcomes from January 2018 onwards. No formal hypotheses are being tested.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Descriptive, retrospective, convenience cohort) **Control Type:** Single-arm (Real-world cohort) **Blinding:** Open-label / N/A **Randomization:** N/A (Convenience cohort)

3.2 Planned Enrollment & Duration Target \$N\$: 200 patients **Number of Sites:** 10 participating Brazilian centers **Estimated Study Start:** 30-Nov-2024 **Estimated Primary Completion:** 31-Dec-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult female or male patients ≥ 18 years old. 2. Confirmed diagnosis of unresectable or metastatic breast cancer (mBC). 3. HR-positive (defined as Estrogen Receptor [ER] $\geq 1\%$) and HER2-negative (defined as IHC 0, 1+ and 2+/ISH-) at the moment first-line treatment was initiated. 4. Received any CDK4/6 inhibitor as first-line treatment since January 2018 and discontinued treatment for any reason (progressed disease, death, or interruption). 5. Patient or next of kin/legal representative willing and able to provide written informed consent according to local regulations (unless waiver granted by local IRB/IEC/EC).

4.2 Exclusion Criteria 1. Patients with recurrent disease with a primary tumor showing a different pattern of receptors (i.e., initial BC other than ER $\geq 1\%$ and HER2 negative). 2. Patients with a concomitant cancer at the time of mBC HR+/HER2-diagnosis (occurring within 5 years of BC diagnosis), except for non-metastatic non-melanoma skin cancers, or in situ or benign neoplasms. 3. Patients who, at the time of data collection, are participating or have participated in an interventional study that remains blinded.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Routine real-world standard of care as prescribed by the treating physician in Brazil. **Route of Administration:** Oral (CDK4/6 inhibitors). **Duration of Treatment:** Retrospective observation of patients who initiated first-line treatment from January 2018 onwards.

5.2 Concomitant & Prohibited Therapy Permitted: All standard-of-care therapies prescribed in the real-world setting. **Prohibited:** Concurrent participation in a blinded interventional clinical trial.

6. Study Timeline & Visit Schedule

(Note: As this is a retrospective observational study, there are no prospective patient visits. Data is obtained via digital chart extraction.)

Visit ID	Window (Days)	Key Activities/Procedures
1	Screening	Identification of eligible digital charts from participating centers; Informed Consent or Waiver.
2	Baseline	Extraction of patient demographics, initial mBC diagnosis details, and ER/HER2 status.
3	Interim	Extraction of first-line CDK4/6i treatment details, treatment duration, and documented tumor response.
4	End of	

Study | N/A | Extraction of discontinuation reasons (progression, death, interruption) and subsequent treatment lines. |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Treatment patterns and clinical outcomes of patients receiving first-line CDK4/6 inhibitors for unresectable or metastatic HR+/HER2- breast cancer. **Measurement Method:** Retrospective data extraction from digital medical charts.

7.2 Secondary & Exploratory Endpoints 1. Real-world progression-free survival and overall survival indicators. 2. Reasons for permanent discontinuation of CDK4/6 inhibitor therapy (progression, toxicity, death).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Retrospective data collection of documented adverse events that led to dose interruptions, reductions, or permanent discontinuation of CDK4/6 inhibitors.

Serious Adverse Events (SAEs): Captured retrospectively as recorded in routine digital health records. **Data Monitoring Committee (DMC):** N/A for retrospective chart review; data anonymization and privacy handled via secure research environment (Vivli.org).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: A convenience cohort of 200 eligible patients treated across the 10 centers. **Sample Size**

Justification: The target sample size is $N = 200$, driven by convenience sampling rather than power calculations, as no formal hypotheses are being tested in this descriptive study. **Handling**

of Missing Data: Dealt with according to standard real-world evidence methodologies (e.g., reported as missing/unknown categories without explicit imputation).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Pharmacoepidemiology Practices (GPP) and local Institutional Review Board (IRB) / Ethics Committee regulations. **Monitoring**

Plan: Source data verification of extracted digital chart records at participating sites. **Audit Trail:** Anonymized individual patient-level data stored and accessed securely via the Vivli.org portal.

11. Study Identifiers & Governance

Primary ID: {{D9673R00053}} Registry Number: {{NCT06676436}}
Sponsor/Lead Organization: {{AstraZeneca}} Study Title:
{{REALNESS}} Official Regulatory Title: {{Real World Outcomes and
Treatment Patterns in Brazilian Patients With Hormone Receptor
Positive / HER2-negative Unresectable or Metastatic Breast
Cancer}}

12. Clinical Framework (Metastatic Breast Cancer
Specific)

Primary Condition: {{Unresectable or Metastatic Breast Cancer
(mBC)}} Diagnosis Threshold: {{HR-positive (ER >= 1%) and HER2-
negative (IHC 0, 1+ and 2+/ISH-)}} Medical Methodology: {{Real-
World Evidence (RWE) Generation}} Optimization Target: {{CDK4/6
inhibitor treatment patterns}}

13. Study Procedures & Methodology

Management Pathway: {{Retrospective digital chart extraction of
Brazilian real-world standard of care}} Education Protocol (HCP):
{{N/A - Retrospective observational study}} Education Protocol
(Patient): {{N/A - Retrospective observational study}} Quality
Audit Mechanism: {{Digital chart source data verification and
anonymized data access via Vivli.org}}

14. Observation & Follow-Up Schedule

Total Duration: {{Retrospective analysis covering treatments from
January 2018 through December 2025}} Onsite Visit Cadence: {{N/A
- Data extracted from digital charts}} Remote Monitoring
Frequency: {{N/A}} Checklist Review Interval: {{N/A}}

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					